Functions

Functions

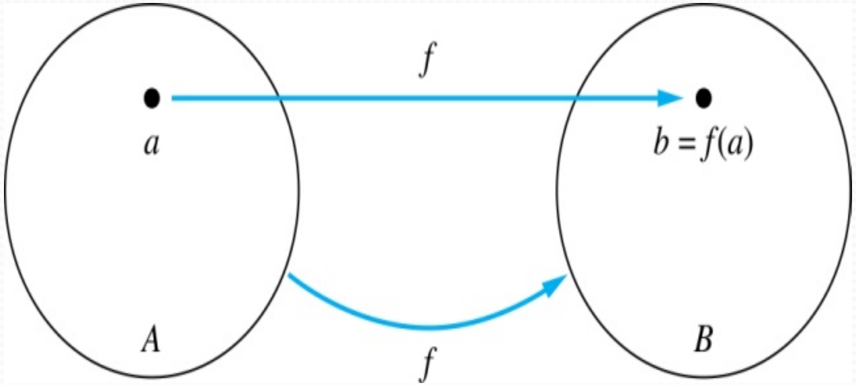
- Functions are an important part of discrete mathematics. A function assigns **exactly one element of a set to each element of the other set.**
- Functions are sometimes called **mappings or transformations.**
- A function , written $f: A \rightarrow B$, is a mathematical relation where each element of a **set A, called the domain**, is associated with a unique element of **another set B, called the codomain** of the function.
- For each element $a \in A$, we associate a unique element $b \in B$. The set of all such$A \rightarrow B$, with (a,b) associations is called function f from A into B, denoted f: used to indicate the mapping f:$a \rightarrow b$, or $f(a)=b$.

Functions

Given a function f: $A \to B$:

- We say f *maps* A **to** B *or*

  f is a *mapping* from A to B.
- A is called the ***domain*** of f.
- B is called the ***codomain*** of f.
- If $f(a) = b$,
  - then **b is called the *image* of a under f.**
  - **a is called the *preimage* of b.**
- The range of f is the set of all images of points in **A** under f. We denote it by $f(A)$.



Example:

- Let:

  $A = \{a,b,c\}$, $B = \{1,2\}$
- Define function $f : A \rightarrow B$ as:
- $f(a) = 1$
- $f(b) = 2$
- $f(c) = 1$

Here:

- **1** is the image of **a**
- **a** is a preimage of **1**

Questions

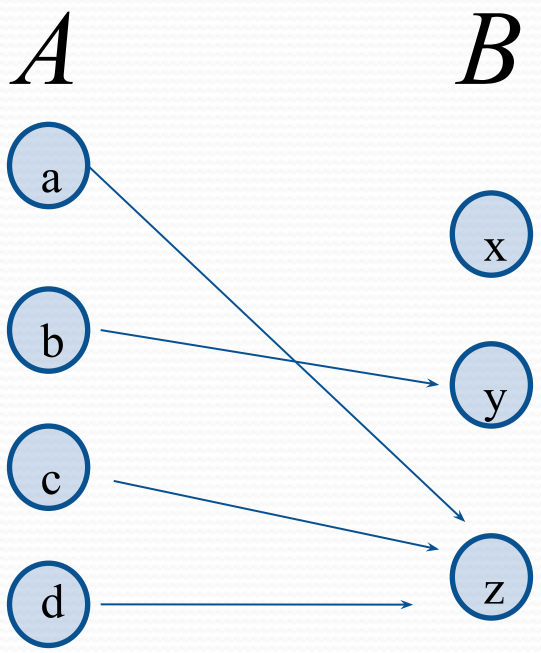
$f(a) = ?$ z

The image of d is ? z

The domain of f is ? A

The codomain of f is ? B

The preimage of y is ? b



$f(A) = ?$ {y,z}

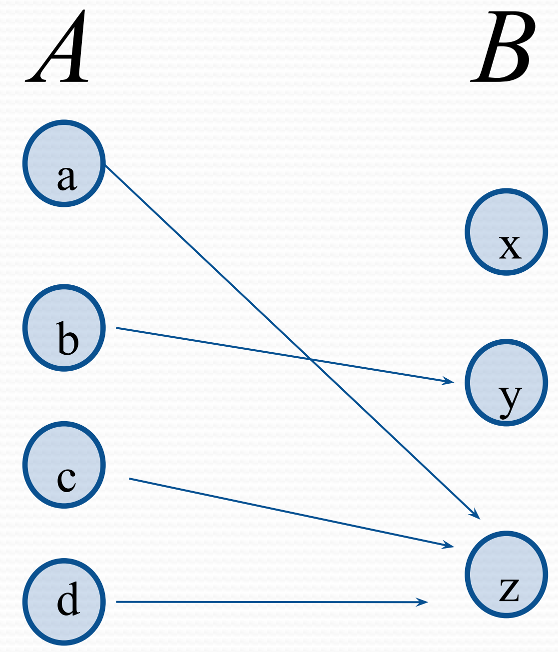
The preimage(s) of z is (are) ? {a,c,d}

Question on Functions

- If $f : A \to B$ then

f {a,b,c,} is ? {y,z}

f {c,d} is ? $\{z\}$

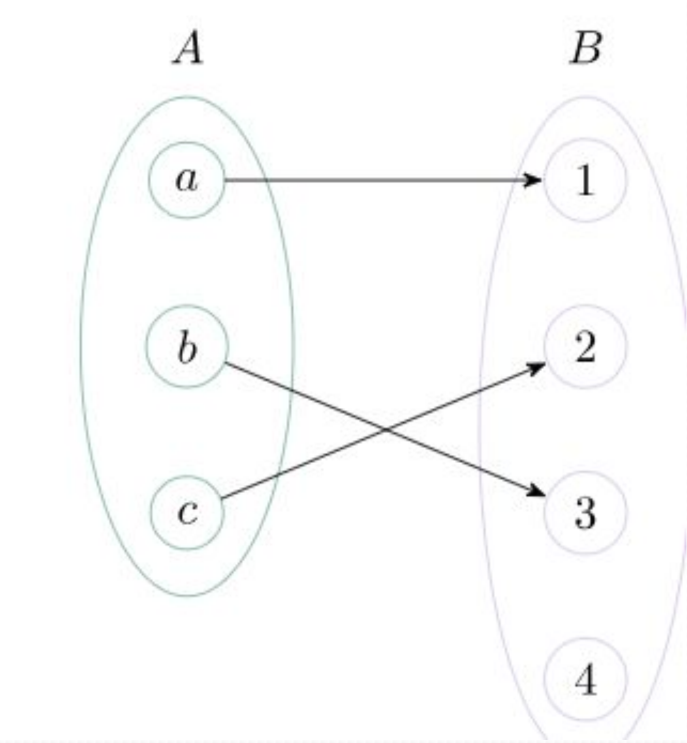


Injective Function

- **An injective (one-to-one) function is a function where each input has a unique output, so no two different inputs produce the same output.**

- **Each input maps to a unique output.**

- **Some elements of the codomain may not mapped to at all.**



Injective Function - Example

- **Example 1:**

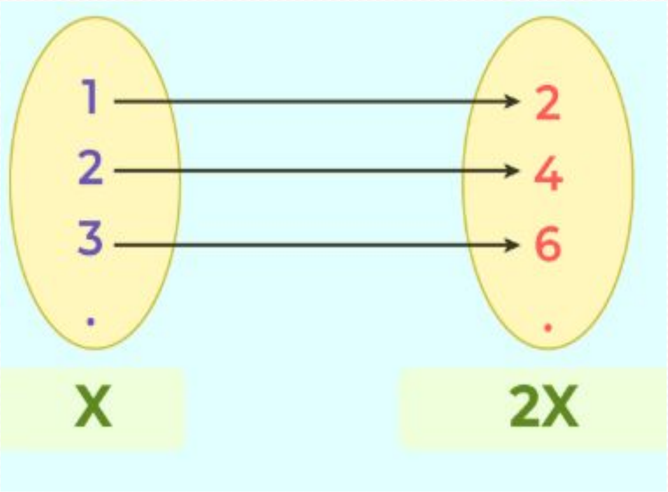
F : {1,2,3,4} → {2,4,6,8,10}

- **Solution:**

f (1) = 2
f (2) = 4
f (3) = 6
f (4) = 8

Injective Function - Example

- **Example 2:** Let's take a function, f(x) = 2x. Is this function injective?
- **Solution:** Yes, f(x) = 2x is indeed an injective function. For every distinct input, you will always get a distinct output.



Surjective Function

- A **surjective (onto) function** is a function where **every element of the codomain is mapped to by at least one element of the domain**—in other words, **no element in the codomain is left unused**.
- **Key point:**
  - Multiple domain elements **can** map to the same codomain element.
  - The only requirement is that **every codomain element has at least one assignment**.

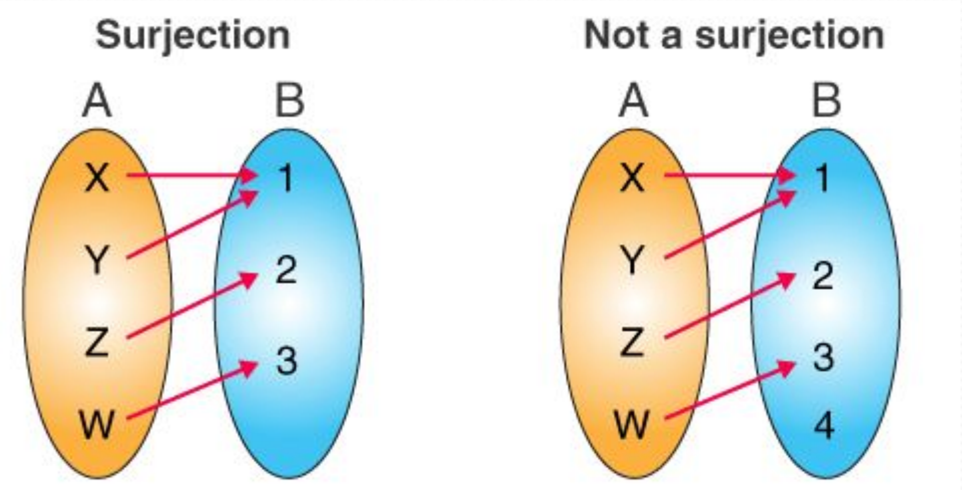


Fig.1 Fig.2

Surjective - Example

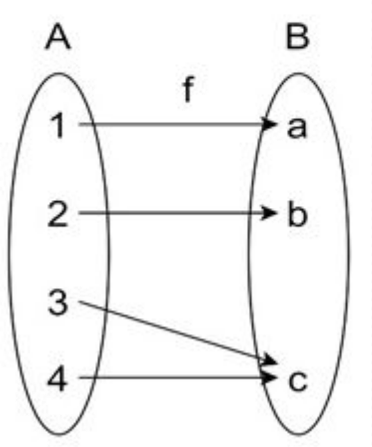
- **Example :** Let us consider the function **f : {1, 2, 3}** → **{a, b, c}** defined by: **f(1) = a, f(2) = b, f(3) = c**

Solution:

Here, every element in the codomain **{a, b, c}** has been "hit" by some element from the domain **{1, 2, 3}.** This means **f** is a surjective function because there are no elements left out in the codomain.

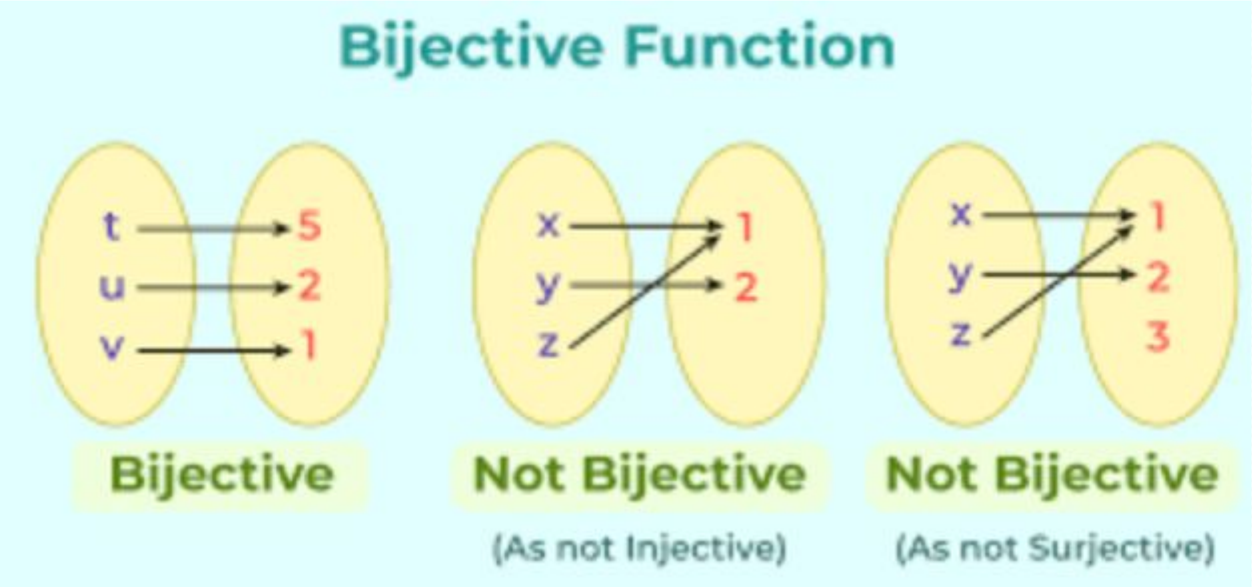
Example 2:

The following mapping presents another **example**



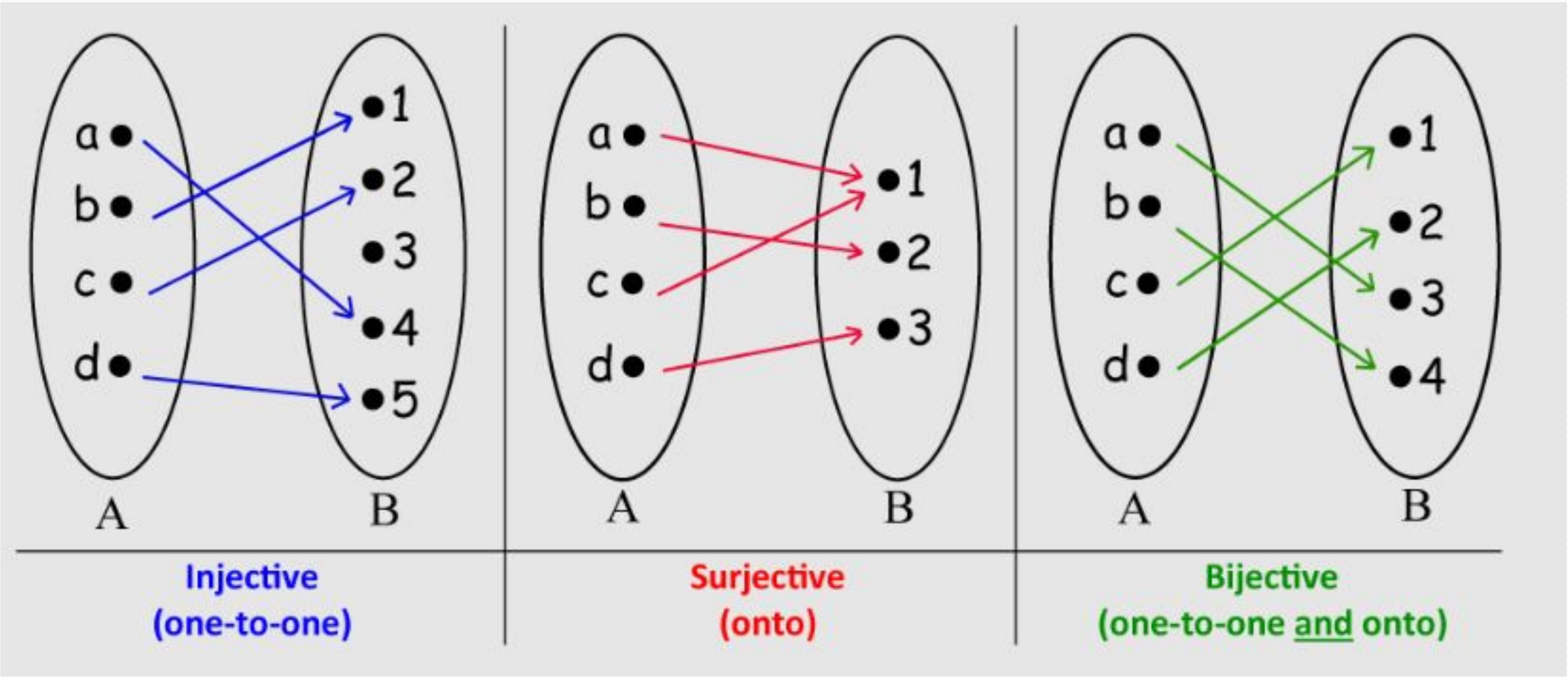
Bijective Functions

- A **bijective function** is a function that is **both injective (one-to-one) and surjective (onto)**.
  This means **each element of the domain maps to a unique element of the codomain, and every element of the codomain is used exactly once**.
  - No repeated outputs (injective)
  - No unused codomain elements (surjective)



Bijective - Example

- **Example 2: Matching Students and ID Numbers?**
- *Suppose you have a set of students (Set A) and a set of student ID numbers (Set B). You want to create a bijective function to pair each student with a unique student ID number.*
- ***Set A (Students):*** *{Alice, Bob, Carol, David}*
  Set B (Student ID Numbers): *{101, 102, 103, 104}*
- **Solution:**
- *You can define a bijective function f as follows:*
- *f(Alice) = 101*
  f(Bob) = 102
  f(Carol) = 103
  f(David) = 104
- *In this example, each student in Set A is matched with a distinct student ID number in Set B, ensuring that no two students share the same ID number (injectivity). Also, every student ID number in Set B has a corresponding student in Set A, so no ID numbers are left unassigned (surjectivity).*


a
b
c
d
1
2
3
4
5
A
B
Injective
(one-to-one)
a
b
c
d
1
2
3
A
B
Surjective
(onto)
a
b
c
d
1
2
3
4
A
B
Bijective
(one-to-one and onto)

Sequences and Summations

Sequences

- **Sequences are ordered lists of elements.**
  - 1, 2, 3, 5, 8
  - 1, 3, 9, 27, 81, …….
  - A sequence is a function whose domain is a subset of the integers, usually{0,1,2,3,…}or {1,2,3,..} and whose codomain is a set *S*.
  - The notation ***an*** **represents the value of the sequence at position *n*.**
  - It is the same as writing ***f*(*n*), where *f* is the function defining the sequence.**
  - Each value ***an*** **is called a term of the sequence.**

EXAMPLE

Let the Function be:

$$an=n^2$$

Then:

- $a1= 1^2 = 1$
- $a2= 2^2= 4$
- $a3=3^2 = 9$
- $a4=4^2 = 16$

So the sequence is:

1,4,9,16,…

Geometric Progression

Definition: A *geometric progression* is a sequence of the form: $a, ar, ar^2, \ldots, ar^n, \ldots$

where the *initial term a* and the *common ratio r* are real numbers.

Real numbers Include:

- Positive numbers (e.g., 2, 5, 3.7)
- Negative numbers (e.g., −1,−4,−0.5)
- Zero (0)
- Whole numbers and fractions (1/2,−3/4)
- Irrational numbers (like $\sqrt{2}, \pi$)

Geometric Progression

Examples:

1. Let $a = 1$ and $r = -1$. Then:

[redacted]

2. Let $a = 2$ and $r = 5$. Then:

[redacted]

3. Let $a = 6$ and $r = 1/3$. Then:

$$\{d_n\} = \{d_0, d_1, d_2, d_3, d_4, \dots\} = \{6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots\}$$

Arithmetic Progression

Definition: A *arithmetic progression* is a sequence of the form:

$$a, a+d, a+2d, \dots, a+nd, \dots$$

where the *initial term a* and the *common difference d* are real numbers.

Examples:

1. Let *a* = −1 and *d* = 4:

[redacted]

2. Let *a* = 7 and *d* = −3:

$$\{t_n\} = \{t_0, t_1, t_2, t_3, t_4, \dots\} = \{7, 4, 1, -2, -5, \dots\}$$

3. Let *a* = 1 and d = 2:

$$\{u_n\} = \{u_0, u_1, u_2, u_3, u_4, \dots\} = \{1, 3, 5, 7, 9, \dots\}$$

Strings

A **string** is a **finite ordered sequence of characters** taken from a fixed set called an **alphabet**.

- Example strings: abcde, hello
- The **length** of a string is the number of characters in it (e.g., abcde has length 5)
- The **empty string**, written as **λ**, has **no characters** and length 0

Strings

A sequence is any ordered list of elements; a string is a sequence specifically made of characters or symbols.

•Sequence: 2, 4, 6, 8 (numbers)

•String: h, e, l, l, o → "hello"

Why strings matter in computer science?

- Words, passwords, and text are strings
- Programs work with strings all the time

Summations

- Sum of the terms $a_m, a_{m+1}, \ldots, a_n$ from the sequence $\{a_n\}$
- The notation:

$$\sum_{j=m}^{n} a_j \quad \textstyle\sum_{j=m}^{n} a_j \quad \textstyle\sum_{m \le j \le n} a_j$$

(read as the sum from j = m to j = n of aj)

represents $a_m + a_{m+1} + \cdots + a_n$

- **The variable *j* is called the *index of summation*.** It runs through all the integers starting with its *lower limit m* and ending with its *upper limit n*.

Summation EXAMPLE

Evaluate the summation:

$$\sum_{i=1}^{5} i$$

Step 1: Write out the terms

This summation means: 1+2+3+4+5

Step 2: Add the terms

1+2=3
3+3=6
6+4=10
10+5=15

Final Answer

$$\sum_{i=1}^{5} i = \boxed{15}$$

Summation EXAMPLE

What is the value of $\sum_{j=1}^{5} j^2$?

Solution: We have

$$
\begin{aligned}
\sum_{j=1}^{5} j^2 &= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 \\
&= 1 + 4 + 9 + 16 + 25 \\
&= 55.
\end{aligned}
$$

Summation EXAMPLE

- Use summation notation to express the sum of the first 100 terms of the sequence {aj},
- The sequence is:

$$a_j = \frac{1}{j}, \quad j = 1, 2, 3, \ldots$$

Summations

SOLUTION:

First 100 terms means:

$$\sum_{j=1}^{100} \frac{1}{j}.$$

- Start at j=1
- End at j=100

- So the terms are:

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{100}$$

Product Notation

- **Product notation** is used when we want to **multiply many terms together**, just like summation is used to add terms.
  - Summation (Σ) → **addition**
  - Product (Π) → **multiplication/ say capital pi**

Product Notation

- Product of the terms $a_m, a_{m+1}, \ldots, a_n$ from the sequence $\{a_n\}$

- The notation:

$$\prod_{j=m}^{n} a_j \qquad \textstyle\prod_{j=m}^{n} a_j \qquad \prod_{m \le j \le n} a_j$$

represents

$$a_m \times a_{m+1} \times \cdots \times a_n$$

Product Notation

Evaluate :

$$\prod_{i=1}^{4} i$$

Step 1: Write out the terms

$1 \times 2 \times 3 \times 4$

Step 2: Multiply

$1 \times 2 = 2$

$2 \times 3 = 6$

$6 \times 4 = 24$

Final Answer

$$\prod_{i=1}^{4} i = \boxed{24}$$